MODEL PAPER UNIT 2

Section-A

<u>Short Answer Questions</u> **– (each of 2 marks)**

2.Explain the working principle of optical fibre.
3. State Gauss's divergence and Stoke's theorem.
4. Explain the physical interpretation of curl.
6. Draw basic structure of optical fibre.
8. Give the statement of Faraday's law.
11. Write down the advantages of optical fibre.
13. Write the formula for poynting vector.
14. What is physical interpretation of divergence?

Section-B

<u>Long Answer based Questions –</u> (each of 4 marks)

1. Draw and explain structure of an optical fibre.
2. Explain EM wave propagation in free space.
3. Drive Poynting Theorem.

Section-C

<u>Long Answer based Questions –</u> (each of 8 marks)

1. Deduce the Maxwell's equations for free space and prove that electromagnetic waves are transverse.
2. Explain the block diagram of optical fiber communication system.